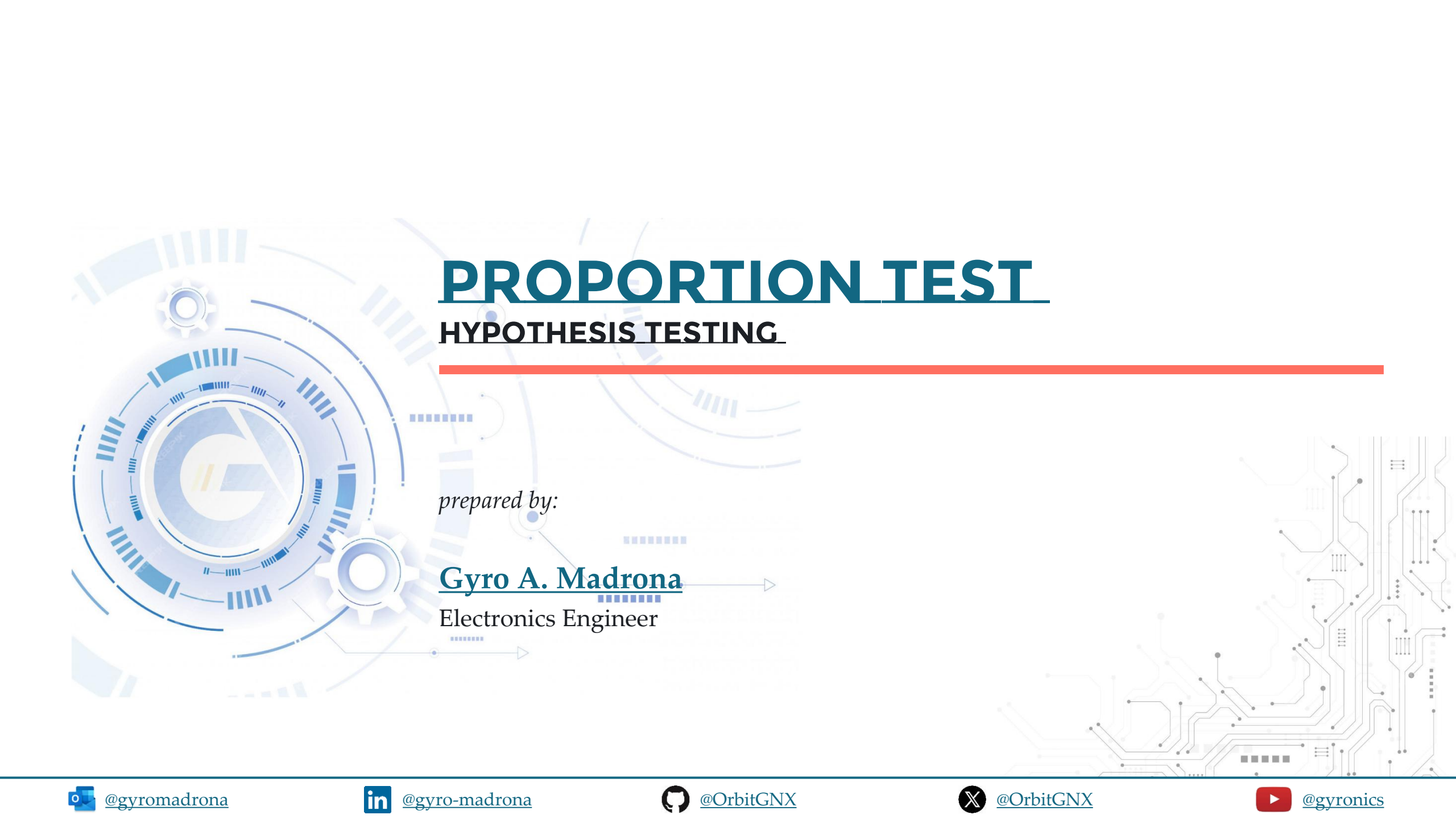




PROPORTION TEST

HYPOTHESIS TESTING

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TOPIC OUTLINE

1-Proportion Test

2-Proportion Test

Analysis of Mean (ANOM)



1-PROPORTION TEST



1-PROPORTION TEST

1-Proportion test is a statistical method used to determine whether a sample proportion differs significantly from a hypothesized population proportion.

Hypothesis

$$H_o: P_1 = P_o$$

$$H_a: P_1 \neq P_o \text{ (p-value} < \alpha \text{)}$$

Assumption

- Discrete data

syntax

```
from scipy import stats

result = stats.binomtest(

    k = number of success,

    n = sample size,

    p = population proportion)

p_value = result.pvalue
```

options

```
alternative = 'two-sided' # default

alternative = 'greater'   # p1 > p0

alternative = 'less'      # p1 < p0
```

1-PROPORTION TEST

1-Proportion test is a statistical method used to determine whether a sample proportion differs significantly from a hypothesized population proportion.

Hypothesis

$$H_o: P_1 = P_o$$

$$H_a: P_1 \neq P_o \text{ (p-value} < \alpha \text{)}$$

Assumption

- Discrete data

syntax

```
from statsmodels.stats.proportion
import proportions_ztest

z_stat, p_value = proportions_ztest(
    count = number of success,
    nobs = sample size,
    value = population proportion)
```

options

```
alternative = 'two-sided' # default
alternative = 'greater'   # p1 > p0
alternative = 'less'      # p1 < p0
```

EXERCISE

In a survey of 1250 people, 600 preferred product A.
Test if this is significantly different from the expected
50% preference.

Solution

Let $\alpha = 0.05$

Hypothesis

$$H_o: P_1 = 0.5$$

$$H_a: P_1 \neq 0.5 \text{ (p-value} < 0.05)$$



2-PROPORTION TEST



2-PROPORTION TEST

2-Proportion test is a statistical method used to determine whether the proportions of two independent groups are significantly different from each other.

Hypothesis

$$H_0: P_1 = P_2$$

$$H_a: P_1 \neq P_2 \text{ (p-value} < \alpha \text{)}$$

Assumption

- Discrete data

syntax

```
from statsmodels.stats.proportion
import proportions_ztest

z_stat, p_value = proportions_ztest(
    count = [success_1, success_2],
    nobs = [size_1, size_2])
```

options

```
alternative = 'two-sided' # default
alternative = 'greater'   # p1 > p2
alternative = 'less'      # p1 < p2
```



EXERCISE

A company produces two types of circuit boards, Board A and Board B. In a quality test:

- 35 out of 150 Board A samples were defective
- 25 out of 120 Board B samples were defective

Is there a significant difference in the defect rates between Board A and Board B at a 5% significance level?

Solution

Let $\alpha = 0.05$

Hypothesis

$$H_0: P_A = P_B$$

$$H_a: P_A \neq P_B \text{ (p-value} < 0.05)$$



ANOM



ANOM

Analysis of Mean (ANOM) is a statistical method used to test whether the means of multiple groups differ significantly from the overall mean.

Hypothesis

$$H_0: P_1 = P_2 = P_3 = \dots P_n$$

$$H_a: \text{at least 1} \neq (\text{p-value} < \alpha)$$

Assumption

- Discrete data

syntax

```
from statsmodels.stats.proportion
import proportions_chisquare

chi_stat, p_value, table =
proportions_chisquare(

    counts = [success_1, success_2,
              success_3, ... success_n],

    nobs = [size_1, size_2,
            size_3, ... size_n])
```



EXERCISE

A company produces two types of circuit boards, Board A and Board B. In a quality test:

- 35 out of 150 Board A samples were defective
- 25 out of 120 Board B samples were defective
- 30 out 85 Board C samples were defective

Is there a significant difference in the defect rates between the boards at a 5% significance level?

Solution

Let $\alpha = 0.05$

Hypothesis

$$H_0: P_A = P_B = P_C$$

H_a : at least 1 board is different (p-value < 0.05)



LABORATORY

